

SDLC Case Study Worksheet

Project Title: E-Learning Platform

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1. Requirements Phase

Functional Requirements:

1. The system shall allow users to create an account and log in to access the platform.
2. The system shall allow users to browse and search courses by category or keyword.
3. The system shall allow users to view course videos and related materials.
4. The system shall allow users to take and submit quizzes.
5. The system shall track and store user progress for each course.

Non-Functional Requirements:

1. The system shall respond to user actions within 2 seconds under normal usage conditions.
2. The UI shall be intuitive and responsive across all major devices.

2. Design Phase

Work Breakdown Structure (WBS) for E-Learning Platform

1. Project Management

1.1 Requirement Analysis

- Gather user and stakeholder requirements
- Define scope and objectives

1.2 Planning & Scheduling

- Define milestones and deliverable
- Assign tasks and timelines

1.3 Quality Assurance

- Conduct testing and validation
- Documentation and review

2. System Design

2.1 Front-End Design

- Design user interface (UI)
- Develop responsive layouts and navigation flow

2.2 Back-End Design

- Design database schema and relationships
- Define server-side architecture and logic flow

2.3 Use Case Diagram

- Model system interactions between users and components
- Define key use cases (User Login, Course Access, Quiz Attempt, etc.)

3. System Development

3.1 Front-End Development

- Implement UI using HTML, CSS, JavaScript/React
- Ensure responsive and accessible design

3.2 Back-End Development

- Implement authentication, database integration
- Develop RESTful APIs for user and course management

3.3 Integration & Testing

- Integrate front-end with back-end services

- Perform functional, usability, and security testing

4. Core Functional Modules

4.1 User Management

- Registration, login, and profile management
- Role-based access (Admin, Instructor, Student)

4.2 Course Management

- Add, update, delete courses
- Upload lessons, videos, and resources

4.3 Assessment & Feedback

- Create quizzes and assignments
- Grading and feedback system

5. Deployment & Maintenance

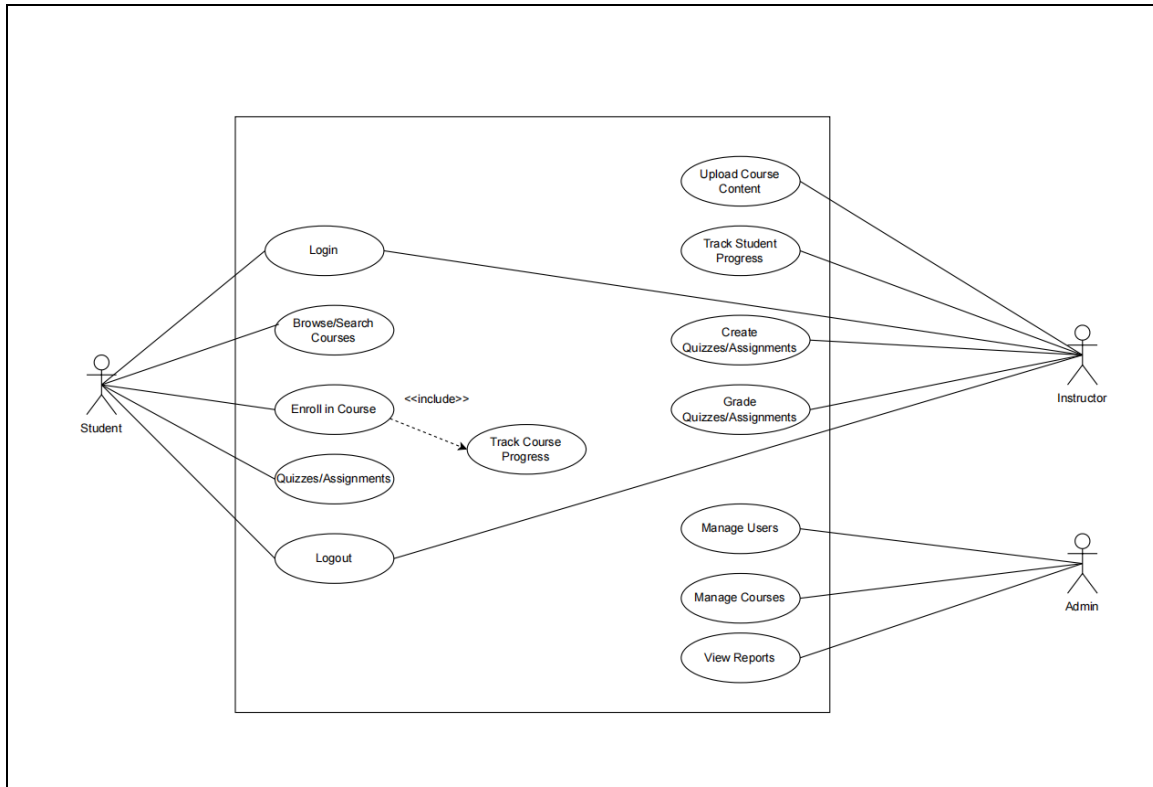
5.1 Hosting & Deployment

- Configure servers and databases
- Deploy on cloud or web hosting platform

5.2 Maintenance & Support

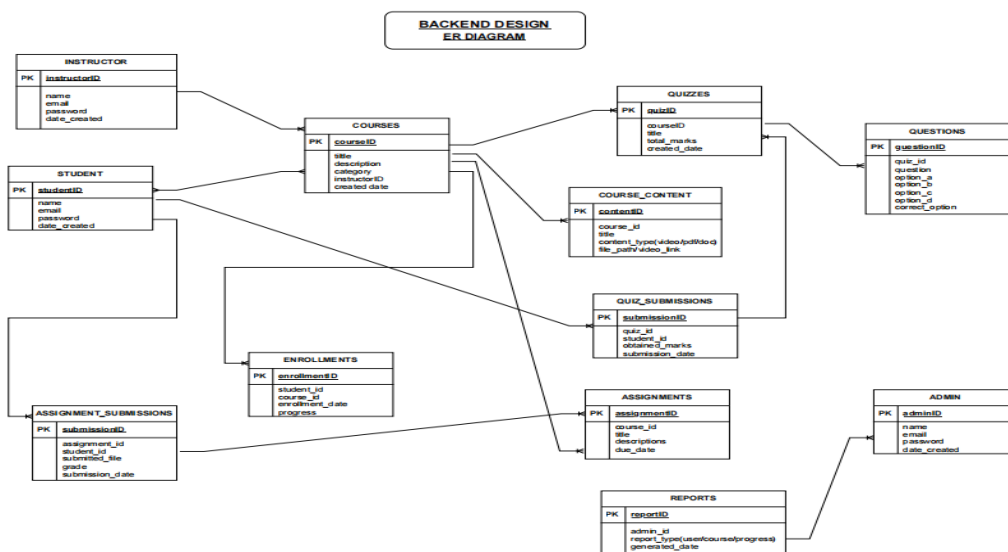
- Monitor system performance and user feedback
- Apply updates, patches, and bug fixes

UML Sketch (Use Case Diagram):



3. Backend Design

Tool Used: Draw.io



4. Development Phase

Pseudo-code:

FUNCTION: EnrollCourse

INPUTS:

userID, courseID

PROCESS (steps):

1. Check if user exists in the system using userID.
2. Check if course exists in the system using courseID.
3. Check if the user is already enrolled in the course.
4. IF user is not enrolled THEN
 - a. Add courseID to user's enrolled courses list
 - b. Add userID to course's enrolled students list
 - c. Generate confirmation message
- ELSE
 - a. Display message "Already Enrolled"
5. Save enrollment data to the database

OUTPUT:

Success or error message (e.g., "Enrollment Successful" or "Already Enrolled")

END FUNCTION

5. Testing Phase

Write 3 test cases.

Test Case ID	Description
TC_ELP_01	Verify that the system allows a user to create a new account and successfully log in.

TC_ELP_02	Verify that users can browse and search courses by keyword or category.
TC_ELP_03	Verify that the system responds to user actions within 2 seconds under normal usage conditions.

Test Case ID	Input(s)	Expected Output	Result (Pass/Fail)
TC_ELP_01	User enters valid details: Username = "Stella123", Email = "Stella@example.com", Password = "Test@123"	System creates the account, displays a success message, and allows login with the same credentials.	Pass
TC_ELP_02	User searches for keyword "Python" in the course catalog.	System displays all available Python-related courses with titles, descriptions, and enrollment options.	Pass
TC_ELP_03	User clicks "Submit Quiz" after answering all questions.	Quiz submission confirmation and score page load within 2 seconds.	Pass

6. Reflection

1. Which SDLC phase was the most challenging? Why?

The design phase was the most challenging because we had to break down the system into detailed parts and plan how everything fits together. It took a lot of effort to ensure that nothing was overlooked.

2. Which SDLC model (Waterfall, Agile) best fits this project? Why?

The Agile model best fits our project because it allows flexibility and regular feedback. Since e-learning platforms often have evolving user needs, Agile supports adapting requirements and improving features throughout the process.

3. How you determine functional and non-functional requirements?

We determined the requirements by considering what users would typically need and expect

from an e-learning platform. We referred to commonly used features in similar systems to identify key functionalities, and we considered general quality standards like usability and performance for non-functional requirements.